FlexControl Knob for Zeus

Version 1.0.1 · by KW4EX · Read Me First

Written for radio operators, not programmers. Every step tells you exactly what to click. You don't need to install any other software.

No warranty — use at your own risk. This add-on is shared free by one radio operator, "as is". There is no warranty of any kind: no promise that it works, that it is safe, or that it suits your equipment or purpose. Amplifiers, radios and antennas can be damaged, and RF can be dangerous. You are responsible for checking that it behaves correctly with your own station, and for operating within your license. The author is not liable for any damage or loss. It is not made, approved or supported by Elecraft, FlexRadio or the Zeus project.

1. What it is

An add-on for **Zeus** that lets the **FlexRadio FlexControl** USB tuning knob control Zeus. It does what the knob did in the KPA1500 Thetis bridge, with Zeus in place of Thetis. It adds a small panel that shows what turning the knob does right now, and what each press does.

2. What it does

- **Turning** tunes VFO A in Zeus's own tuning step. A press can switch the knob to AF volume, AGC gain, VFO B, RIT, XIT, tuning step or TX drive. The three button lights on the knob show which.
- **12 presses** (the knob press and buttons 1–3, each as a tap, double tap or long press). Each runs a function you choose in the panel's ⚙ **Settings**.

Press	Tap	Double tap	Long press
Knob	AF volume ⇄ VFO A	Mute	Back to VFO A
Button 1	AF volume ⇄ VFO A	AGC gain ⇄ VFO A	Mute
Button 2	Noise reduction	Tuning step larger	Tuning step smaller
Button 3	Safe TUNE / stop	Band up	Band down

These are the starting settings. Change any of them in ⚙ Settings. Other choices include Mute, VFO lock, Split, RIT and XIT, NR / NB / ANF, MOX (which transmits!), amplifier OPERATE / STANDBY, and any Zeus TCI command.

3. What you need

Item	Details
Zeus	Installed and working, on a version that has **Features** with a **Community** section.
FlexControl knob	Plugged into the **computer that runs Zeus** with its USB cable.
Zeus TCI turned on	A setting inside Zeus (normally port **40001**). Without it, only VFO A tuning works.
Optional: KPA1500 feature	The Elecraft KPA1500 feature for Zeus, if you want the knob's safe TUNE and amplifier presses.
Other software	**None.**

Only one program can use the knob at a time. Close DDUtil, the KPA1500 Thetis bridge, or any other program that uses the knob before starting Zeus.

4. What's in this folder

Folder or file	What it is
READ ME FIRST ...pdf	This guide.
1 Windows PC	The install file for Windows.
2 Mac - Apple silicon (M1-M4)	The install file for newer Macs.
3 Mac - Intel	The install file for older Intel Macs.
4 Linux PC (64-bit)	The install file for an ordinary Linux PC.
5 Raspberry Pi (64-bit)	The install file for a Raspberry Pi 4 or 5.

Each numbered folder holds one file ending in `.zip`. That file **is the add-on**. Pick the folder that matches the computer running Zeus (on a Mac: Apple menu › **About This Mac** shows "Apple M1...M4" or "Intel"; on a Raspberry Pi type `uname -m` in Terminal, it must say `aarch64`).

Do not unzip (extract) the .zip file. Give it to Zeus exactly as it is.

5. Install (all computers)

1. Close any other program that uses the knob (see Part 3). Plug the knob in.
2. Copy the right numbered folder onto the computer that runs Zeus (a USB stick or your home network works; on Windows it is already here under **C: › RadioScripts**).
3. Start **Zeus**. You don't need to remove an older version first; installing a new one replaces it.
4. Open **Features** › **Community** › **Install local feature**, pick the `.zip` file and click **Open**.
5. If Zeus asks to restart, close it and start it again.
6. Add the panel: **Add Panel › Controls › FlexControl Knob**.
7. In Zeus's own settings, turn **TCI** on (port **40001**).
8. Turn the knob. VFO A should tune. If it doesn't, open the panel's ⚙ **Settings** and check the **Port** (Part 6).

Linux and Raspberry Pi: serial ports are open only to members of the **dialout** group. Once, in Terminal, type `sudo usermod -aG dialout $USER`, enter your password, then log out and back in.

Mac: if the Mac asks to let Zeus use the network, click **Allow**.

6. First-time settings

- **Port: Auto** finds the knob by its USB id (Windows and Linux). On a Mac, Auto works when the knob is the only USB modem; otherwise choose the port that starts `/dev/cu.usbmodem`.
- **Presses:** in ⚙ Settings, choose what each of the 12 presses does.
- **Knob lights** (buttons 1-2-3): VFO A ○○○ · AF volume ●○○ · AGC gain ○●○ · RIT ○○● · XIT ○●● · VFO B ●○● · Tuning step ●●○ · TX drive ●●●

7. Safety rules built in

- **TUNE** puts the KPA1500 in STANDBY first and waits for it to confirm. It is refused if the amp can't be reached, if the amp doesn't confirm, if the "No antenna?" warning is showing, or while transmitting. TUNE stops by itself after 20 seconds. Pressing it again stops it sooner.
- **MOX** (transmit) is refused while the "No antenna?" warning is showing.
- **Band changes** are refused while transmitting.
- **Tuning** is ignored while the VFO is locked in Zeus.

8. Updating to a newer version

Each new version comes in its own folder with the version number in its name. In Zeus use **Features › Community › Install local feature** and pick the new `.zip`. It replaces the old version.

Close and restart Zeus. The newest version is always at `github.com/kw4ex/zeus-amp/tree/builds/flexknob/latest`.

9. If something goes wrong

What you see	What to do
Zeus won't install the file	Pick the `.zip` *file*, not a folder, for the right computer type. Don't unzip it.
Knob does nothing	Close DDUtil, the Thetis bridge or any other program using the knob. In ⚙ Settings, check the Port and press **Refresh**. On Linux, do the dialout step (Part 5).
Only VFO A tuning works	Turn **TCI** on in Zeus (port 40001).
TUNE press is refused	The KPA1500 feature must be installed and connected, and the amp must reach STANDBY. Check the message on the panel.

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